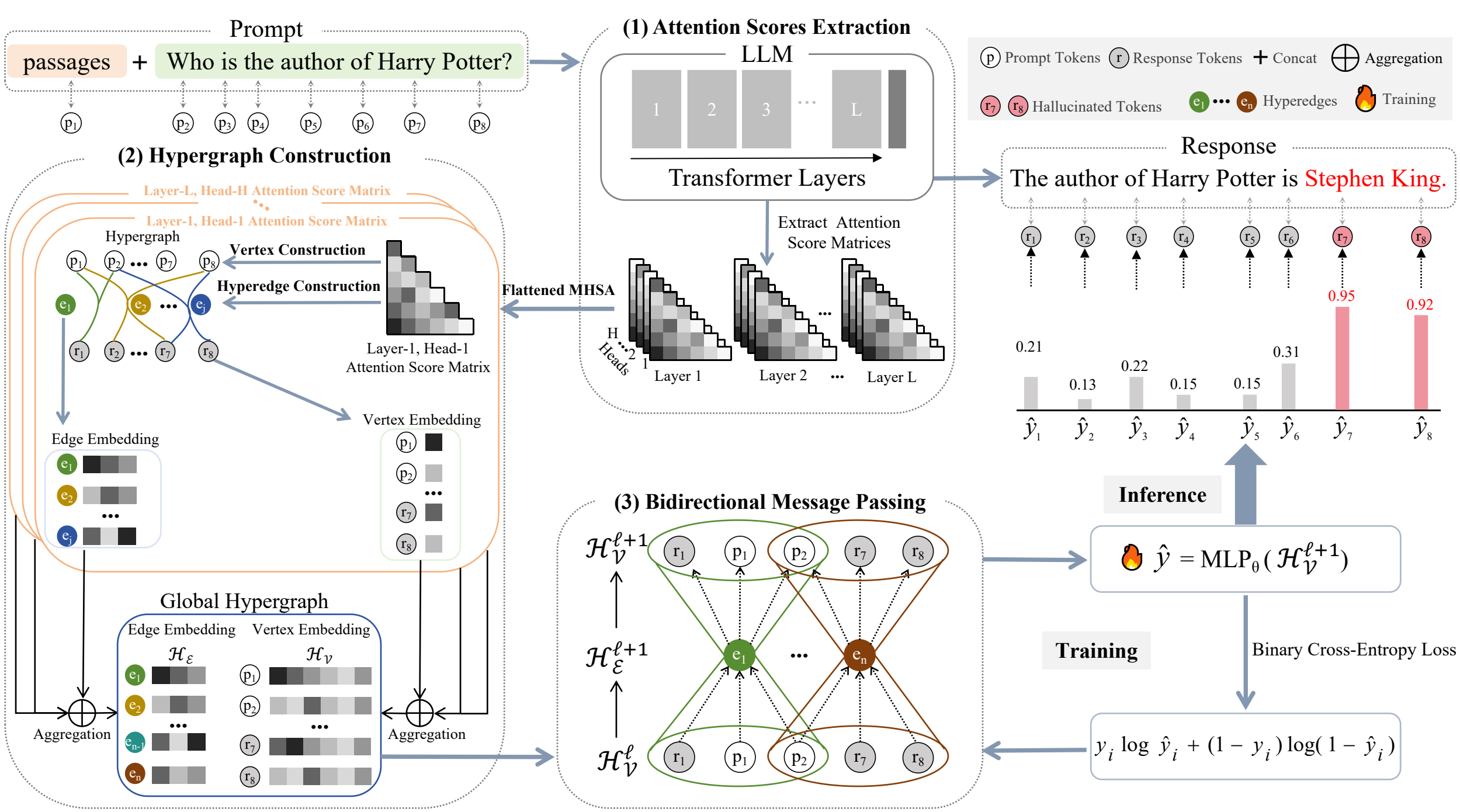

Prompt
passages + Who is the author of Harry Potter?
(1) Attention Scores Extraction
LLM
Transformer Layers
Extract Attention Score Matrices
Flattened MHSA
Layer 1
Layer 2
Layer L
Heads
(2) Hypergraph Construction
Layer-L, Head-H Attention Score Matrix
Layer-1, Head-1 Attention Score Matrix
Hypergraph
Vertex Construction
Hyperedge Construction
Layer-1, Head-1 Attention Score Matrix
Edge Embedding
Vertex Embedding
Global Hypergraph
Aggregation
Aggregation
(3) Bidirectional Message Passing
Prompt Tokens
Response Tokens
Concat
Aggregation
Hallucinated Tokens
Hyperedges
Training
Response
The author of Harry Potter is Stephen King.
0.21
0.13
0.22
0.15
0.15
0.31
0.95
0.92
Inference
Training
Binary Cross-Entropy Loss